MARITIME POWER RESEARCH · CYBER DEFENSE · STRATEGIC STUDIES

Cyber power as a structural variable of maritime power

A normalized multiplicative model for multidomain assessment of maritime power

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$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$n_{eff} = 3 + \gamma + \alpha$ · $MP_{norm} \in (0, 1]$ · weighted geometric mean · keeps the classical core MP = MI × NP

FROM THE CLASSICAL TO THE EXTENDED FORMULA

The classical formula: sound, but incomplete. MP = MI × NP (Solís Oyarzún, 1997; Uribe Cáceres et al., 2016) does not include cyberspace, strategic will or the security of the environment as structural variables.

The proposal: update it, not replace it. Three structural variables and a normalization that makes the result interpretable. Existing measures are additive; this one is non-compensatory.

THE FIVE VARIABLES – TWO LEVELS

MI
Maritime interests

$MI = \Sigma\ w_i \cdot MI_i$

Weighted sub-index. Colombia: 18 interests (CCO, 2021).

NP
Naval power

$NP = F \times P \times MDA^{\mu}$

Force, position and maritime domain awareness.

SW
Strategic will

$(PW \cdot IW \cdot SocW)^{1/\Sigma w}$

Multiplier of the system: political, institutional and social.

MS
Maritime security

$(IC \cdot TE \cdot BC)^{1/\Sigma a}$

Incidence of threats: conflicts, terrorism, blue crime.

CP
Cyber power

$\min\{1, (\beta_1 CS_m + \beta_2 CD_m + \beta_3 RC_m)(1+\delta \cdot CDip)\}$

Cybersecurity, strategic cyber defense, recovery + cyber diplomacy.

INTERPRETATION SCALE – COLOMBIA AT THE MODERATE LEVEL

Critical	Low	Moderate	Significant	High
0.00 – 0.20	0.21 – 0.40	0.41 – 0.60	0.61 – 0.80	0.81 – 1.00

▲ **Colombia: MP_{norm} = 0.455** – Moderate level. Only MS comes from an external source (rank 75 of 99, Vargas-Suárez et al., 2021 → 0.24); the other values are illustrative.

ROBUSTNESS

With inputs ± 0.10, Moderate in 96.6% of the draws; with ± 0.20, in 78.1% (Low, 21.0%). MS and γ dominate the variance.

Where to invest? Sensitivity is not enough: costs are needed

Proposition 1. $MP_{norm} = G^{1-w} \cdot CP^{w}$, $w = \alpha / n_{eff}$: raising CP always raises the index; raising α only if CP > G. α is a weight, not a capability.

Marginal return (Colombia). MS 0.312 · SW 0.223 · NP 0.195 · MI 0.178 · CP 0.147.

Corollary 1. Cyber power would be the most efficient vector only if raising the others cost at least 1.21× (MI), 1.32× (NP), 1.52× (SW), 2.12× (MS).

FIVE CONTRIBUTIONS

1 · Structural CP
With explicit operational boundaries.

2 · System-wide SW
Multiplier of all of MP.

3 · Disaggregated MS
Incidence of threats, weight γ.

4 · Normalization
Five-level scale; verification.

5 · MDA ≠ consciousness
Technical versus cultural.

Agenda: National Maritime Power Index with a calibration and external-validation protocol · regional extension · dynamic modeling · **Simulator:** diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html